```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

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```
uploaded = files.upload()

video_path = next(iter(uploaded))
print("Video uploaded:", video_path)
```

```
Choose Files No file chosen    Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving vidssave.com switzerland 4k _ switzerland 4k video _ switzerland 480p.mp4 to vidssave.com switzerland 4k _ switzerlar
Video uploaded: vidssave.com switzerland 4k _ switzerland 4k video _ switzerland 480p.mp4
```

```
cap = cv2.VideoCapture(video_path)

ret, frame1 = cap.read()
ret, frame2 = cap.read()

if not ret:
    print("Could not read video")
else:
    print("Video loaded successfully")

cap.release()
```

```
Video loaded successfully
```

```
gray1 = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(frame2, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.imshow(gray1, cmap='gray')
plt.title("Frame 1")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(gray2, cmap='gray')
plt.title("Frame 2")
plt.axis("off")

plt.show()
```




```
# Detect feature points
features = cv2.goodFeaturesToTrack(
    gray1,
    maxCorners=100,
    qualityLevel=0.3,
    minDistance=7,
    blockSize=7
)

# Lucas-Kanade parameters
lk_params = dict(
    winSize=(15, 15),
```

```
    maxLevel=2,
    criteria=(cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 10, 0.03)
)

# Calculate optical flow
next_points, status, error = cv2.calcOpticalFlowPyrLK(
    gray1, gray2, features, None, **lk_params
)

# Select good points
good_old = features[status == 1]
good_new = next_points[status == 1]

# Draw motion vectors
output = frame2.copy()

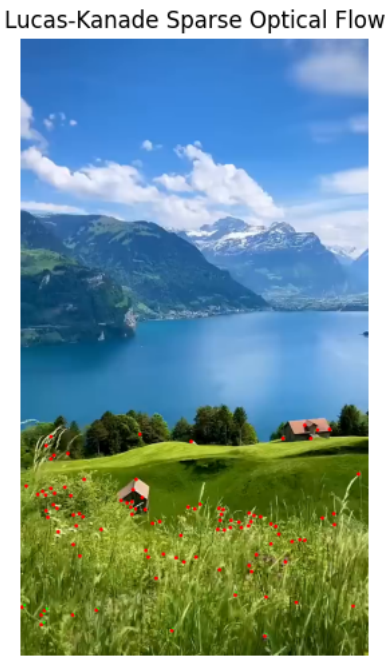
for old, new in zip(good_old, good_new):
    x1, y1 = old.ravel()
    x2, y2 = new.ravel()

    cv2.arrowedLine(
        output,
        (int(x1), int(y1)),
        (int(x2), int(y2)),
        (0, 255, 0),
        2
    )

    cv2.circle(
        output,
        (int(x2), int(y2)),
        3,
        (0, 0, 255),
        -1
    )

plt.figure(figsize=(12, 6))
plt.imshow(cv2.cvtColor(output, cv2.COLOR_BGR2RGB))
plt.title("Lucas-Kanade Sparse Optical Flow")
plt.axis("off")
plt.show()
```

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```
flow = cv2.calcOpticalFlowFarneback(
    gray1,
    gray2,
    None,
    0.5,
    3,
    15,
    3,
    5,
    1.2,
    0
)
```

```
# Convert flow to magnitude and angle
magnitude, angle = cv2.cartToPolar(
    flow[..., 0],
    flow[..., 1]
)
```

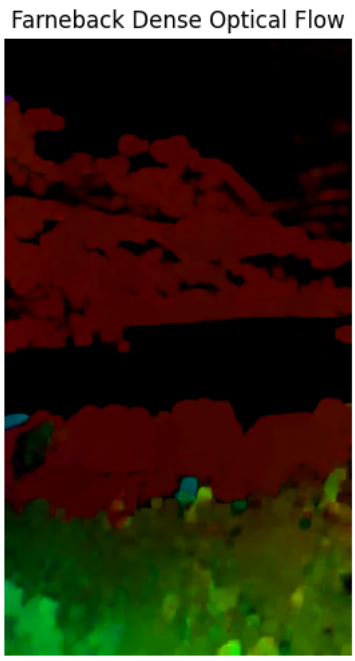
Add text cell

```
# Create HSV image
hsv = np.zeros_like(frame1)
hsv[..., 1] = 255

hsv[..., 0] = angle * 180 / np.pi / 2
hsv[..., 2] = cv2.normalize(
    magnitude,
    None,
    0,
    255,
    cv2.NORM_MINMAX
)

dense_flow = cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB)

plt.figure(figsize=(12, 6))
plt.imshow(dense_flow)
plt.title("Farneback Dense Optical Flow")
plt.axis("off")
plt.show()
```




```
print("Optical Flow Algorithm Comparison")
print("---------------------------------")
print("Lucas-Kanade : Sparse Optical Flow")
print("Farneback     : Dense Optical Flow")
print()
print("Lucas-Kanade tracks selected feature points.")
print("Farneback estimates motion for almost every pixel.")
print("Lucas-Kanade generally requires less computation.")
print("Farneback provides more detailed motion information.")
```

```
Optical Flow Algorithm Comparison
---------------------------------
Lucas-Kanade : Sparse Optical Flow
Farneback     : Dense Optical Flow

Lucas-Kanade tracks selected feature points.
Farneback estimates motion for almost every pixel.
Lucas-Kanade generally requires less computation.
Farneback provides more detailed motion information.
```

Experiment 7: Motion Estimation using Optical Flow Algorithms

Questions and Answers

1. What is Optical Flow? How is it used in computer vision?

Answer: Optical Flow is a computer vision technique used to estimate the apparent motion of objects, surfaces, or camera movement between consecutive frames of a video sequence. It is used to analyze motion patterns and is useful in applications such as autonomous driving, video surveillance, robotics, gesture recognition, sports analytics, and augmented reality.

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2. Explain the working principle of the Lucas-Kanade Optical Flow algorithm.

Answer: Lucas-Kanade is a Sparse Optical Flow algorithm. It first selects important feature points in an image and then tracks these points in the next frame. It assumes that the motion within a small neighborhood around a feature point is approximately similar. By comparing the position of the feature points in consecutive frames, the algorithm calculates their displacement and motion vectors.

3. What is Dense Optical Flow? How does it differ from Sparse Optical Flow?

Answer: Dense Optical Flow estimates the motion of almost every pixel in an image. In contrast, Sparse Optical Flow estimates motion only for selected feature points.

Sparse Optical Flow is computationally less expensive and is suitable for feature or object tracking. Dense Optical Flow provides a detailed motion field but requires more computational processing. Lucas-Kanade is commonly used for Sparse Optical Flow, while Farneback is used for Dense Optical Flow.

4. Compare the Lucas-Kanade and Farneback optical flow algorithms.

Answer:

Feature	Lucas-Kanade	Farneback
Type	Sparse Optical Flow	Dense Optical Flow
Motion estimation	Selected feature points	Almost every pixel
Computational cost	Lower	Higher
Output	Motion vectors at feature points	Dense motion field
Main use	Feature and object tracking	Detailed motion analysis

Lucas-Kanade is useful when only important feature points need to be tracked, whereas Farneback is useful when motion information across the image is required.

5. What assumptions are made while computing optical flow?

Answer: The main assumptions made while computing optical flow are:

1. The brightness or intensity of a moving point remains approximately constant.
2. The motion between consecutive frames is relatively small.
3. Neighboring pixels have approximately similar motion.
4. Image intensity changes smoothly enough to estimate motion.

6. What factors can affect the accuracy of optical flow estimation?

Answer: The accuracy of optical flow estimation can be affected by:

1. Object speed.
2. Lighting conditions.
3. Camera motion.
4. Image noise.
5. Motion blur.
6. Occlusion of objects.
7. Large displacement between frames.
8. Lack of texture in some regions.

7. Mention five real-world applications of optical flow in computer vision.

Answer: Five real-world applications of optical flow are:

1. Autonomous driving.
2. Video surveillance.
3. Robotics.
4. Gesture recognition.
5. Sports analytics.

8. Why are grayscale images generally used for optical flow computation?

Answer: Grayscale images are generally used because they contain a single intensity value for each pixel instead of three color channels. Optical flow mainly depends on changes in image intensity and image gradients. Using grayscale images simplifies the computation and reduces processing requirements.

9. What are the limitations of optical flow algorithms in real-world environments?

Answer: The limitations of optical flow algorithms include:

1. They can be sensitive to changes in lighting.
2. Very fast motion Add text cell cult to estimate accurately.
3. Motion blur can reduce accuracy.
4. Occluded objects are difficult to track.
5. Camera movement can create large apparent motion.
6. Textureless regions provide fewer useful features.
7. Dense optical flow can require high computational resources.

10. How does optical flow contribute to applications such as autonomous driving, video surveillance, and action recognition?

Answer: Optical flow provides information about the movement of objects and regions between consecutive video frames.

- **Autonomous Driving:** Optical flow can help estimate the movement of surrounding objects and camera motion.
- **Video Surveillance:** It can be used to detect and track moving people or objects.
- **Action Recognition:** Motion patterns between consecutive frames can help identify different human activities.

Therefore, optical flow is an important technique for understanding motion in dynamic video scenes.

```
Start coding or generate with AI.
```